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AI dependence and cognition studies

AI Dependence and Human Cognition: Impacts, Evidence, and Design for Independent Thought (2020–2026)

Quick Reference

Key Findings Table

Cognitive Domain	AI Impact (2020–2026)	Evidence Strength	Key Moderators/Interventions	Limitations/Gaps
Memory	Risk of cognitive atrophy, reduced retention, hippocampal disengagement	High	Metacognitive scaffolding, AI literacy	Few longitudinal/neuroimaging studies
Critical Thinking	Dual effect: can enhance or diminish, depending on pedagogy and AI literacy	High	Teacher mediation, inquiry-based learning	Overreliance, lack of causal evidence
Reasoning	Improved efficiency, but automation bias and reduced adaptability	Moderate	Human-in-the-loop, algorithmic audits	Cross-sectional bias, cultural gaps
Problem-Solving	Enhanced by adaptive AI, but risk of cognitive offloading	High	Adaptive scaffolding, feedback cycles	Underexplored in marginalized groups
Decision-Making	Increased accuracy, but automation bias and reduced reflective judgment	Moderate	Metacognitive prompts, transparency	Limited cross-cultural/long-term data
Attention/Self-Regulation	Declines with overuse; improved by adaptive AI and SRL training	Moderate	Digital well-being pedagogy, SRL frameworks	Short-term focus, lack of diversity

Direct Answer

The reviewed studies from 2020 to 2026 indicate that while AI can enhance learning efficiency, creativity, and engagement, there is also clear evidence of risks associated with overreliance—including memory decline, reduced critical thinking, and cognitive offloading. Methodologies predominantly employing cross-sectional surveys and qualitative analyses suggest the need for more longitudinal and experimental designs to establish causal links. Although existing AI solutions like adaptive tutoring systems, generative AI models (e.g., ChatGPT), and intelligent feedback tools can support learning, their design should integrate metacognitive scaffolding, human-centered principles, and ethical frameworks to foster independent thought. Moreover, interventions like interactive pedagogical strategies, AI literacy training, and teacher mediation have been identified as vital in mitigating negative impacts while promoting higher-order cognitive skills. Future AI designs should therefore prioritize cognitive alignment, transparency, and equitable access, ensuring that AI serves as an augmentation to human intelligence rather than a substitute for it [1] [2] [3].

Study Scope

- **Time Period:** 2020–2026
- **Disciplines:** Education, cognitive neuroscience, psychology, computer science, ethics, health sciences
- **Methods:** Systematic reviews, cross-sectional and longitudinal surveys, experimental and quasi-experimental designs, neuroimaging, mixed-methods, algorithmic audits

Assumptions & Limitations

- Heavy reliance on self-reported and cross-sectional data; limited causal inference
- Underrepresentation of marginalized populations and early learners
- Scarcity of longitudinal, neuroimaging, and cross-cultural studies
- Inconsistent operationalization of “AI dependence” and cognitive constructs
- Limited integration of neuropsychological and quantum-like frameworks

Suggested Further Research

- Longitudinal, theory-driven studies on AI’s causal impact on cognition
- Cross-cultural and interdisciplinary research, especially in underrepresented groups
- Integration of neuroimaging, algorithmic audits, and behavioral data
- Development and empirical validation of unified theoretical frameworks (e.g., neuropsychological, quantum-like, socio-technical)
- Standardized protocols for evaluating AI’s cognitive and ethical impacts

1. Introduction

The rapid integration of artificial intelligence (AI) into educational, professional, and everyday contexts has transformed how individuals learn, reason, and make decisions. As AI systems—from generative models like ChatGPT to adaptive tutoring platforms—become ubiquitous, understanding their impact on core cognitive functions is critical. This review synthesizes peer-reviewed research from 2020 to 2026, focusing on the nuanced effects of AI dependence on memory, critical thinking, reasoning, problem-solving, and decision-making, and evaluates solutions and design principles that support learning without replacing independent thought [1] [4] [5].

Scope and Rationale

This report draws on empirical and theoretical studies across disciplines, highlighting both the promise and perils of AI-mediated cognition. It addresses methodological strengths and gaps, evaluates interventions, and proposes actionable design principles for AI systems that foster, rather than supplant, human cognitive skills [1] [4] [5].

2. Impact of AI Dependence on Cognitive Functions

Memory and Cognitive Offloading

- **Cognitive Atrophy:** Prolonged AI reliance leads to reduced memory retention and cognitive atrophy, attributed to underuse of neural circuits and habitual cognitive offloading [1] [6].
- **Neural Mechanisms:** Neuroimaging and computational models reveal that AI dependence alters activity in prefrontal-parietal and hippocampal networks, reducing neuroplasticity and synaptic engagement, especially in adolescents whose executive functions are still developing [1] [7] [8].
- **Hippocampal Systems:** Reduced cognitive engagement from AI offloading may impair hippocampal-dependent memory consolidation, with long-term risks for memory encoding and retrieval [6] [8].

Synthesis: While AI can support memory through externalization and reminders, unreflective dependence risks neural and cognitive decline, particularly in vulnerable age groups. Balanced human-AI collaboration and cognitive training are essential to preserve memory systems [1] [6].

Critical Thinking and Creativity

- **Dual Impact:** AI can both enhance and diminish critical thinking and creativity, depending on integration and user literacy. Structured, scaffolded AI use supports higher-order thinking, while unregulated use fosters cognitive laziness and superficial engagement [9] [10] [11] [12].
- **Educational Level Differences:** AI's positive effects on creativity are more pronounced in primary and secondary education, while higher education students risk diminished critical thinking if AI is used as a substitute rather than a collaborator [13] [14].
- **Mediators:** Digital literacy, motivation, and academic support moderate AI's impact on creativity and critical thinking. Information literacy can buffer negative effects, but cognitive fatigue remains a risk [15] [16].

Synthesis: Intentional pedagogical strategies, AI literacy, and metacognitive scaffolding are crucial to harness AI's creative potential while safeguarding critical thinking [12] [17].

Reasoning, Problem-Solving, and Decision-Making

- **Efficiency vs. Bias:** AI tools improve efficiency and accuracy in reasoning and problem-solving but can induce automation bias, reduce heuristic adaptability, and undermine reflective judgment [18] [19] [20] [21].
- **Executive Function:** Overreliance on AI weakens executive functions such as planning, sequencing, and self-regulation, with cross-cultural studies showing variability in cognitive engagement and susceptibility to bias [20] [22].
- **Age and Culture:** Adolescents and younger users are more vulnerable to cognitive atrophy and automation bias, while cultural factors influence critical thinking disposition and AI engagement [21] [23].

Synthesis: Human oversight, algorithmic audits, and culturally responsive pedagogy are needed to mitigate automation bias and preserve reasoning skills [20] [21].

Sustained Attention and Self-Regulation

- **Attention Decline:** Excessive AI use is linked to reduced sustained attention and self-regulation, undermining planning and independent judgment [24] [25].
- **Interventions:** Adaptive tutoring, metacognitive training, and digital well-being pedagogy improve attention and self-regulation, especially when integrated with self-regulated learning (SRL) frameworks [24] [26].

Synthesis: AI can both challenge and support attention/self-regulation; the key is adaptive, scaffolded use that promotes self-monitoring and goal-setting [24] [25].

3. Methodologies and Limitations in AI and Cognition Research

Research Methodologies

- **Systematic Reviews & Mixed-Methods:** Most studies use PRISMA-guided reviews, bibliometric analyses, and mixed-methods (quantitative surveys, qualitative interviews, network analysis) [27] [28] [29] [30].
- **Experimental Designs:** Few longitudinal or experimental studies exist; most are cross-sectional, limiting causal inference [31] [32].
- **Neuroimaging:** MRI, PET, fMRI, and EEG are used to assess neural correlates of AI use, but integration with behavioral data is rare [29] [33].

Longitudinal and Experimental Evidence

- **Sparse Long-Term Data:** Only a handful of multi-wave or year-long studies track AI dependence and cognitive outcomes, often focusing on mediators like self-efficacy and moderators like gender [31] [34].
- **Causal Inference:** Single-case experimental designs and emulated target trials are proposed to strengthen causal claims, but are not yet widely adopted [32] [35].

Mixed-Method Approaches and Metacognitive Shifts

- **Metacognitive Insights:** Mixed-methods reveal that AI-based feedback improves metacognitive planning and monitoring, but may increase dependency unless paired with teacher-guided reflection [30] [36].
- **Adaptive Scaffolding:** Deep reinforcement learning and adaptive AI interventions outperform static approaches in supporting metacognitive growth [37].

Limitations and Future Directions

- **Reproducibility & Complexity:** Black-box AI models, lack of transparency, and computational complexity hinder validation and trust [38] [39].
- **Underrepresentation:** Marginalized groups, early learners, and non-STEM disciplines are underrepresented [38] [39].
- **Integration Gaps:** Insufficient integration of neuroimaging, algorithmic audits, and behavioral data limits understanding of long-term effects [40] [41].

Synthesis: Methodological innovation—especially longitudinal, interdisciplinary, and mixed-method designs—is urgently needed to advance the field [3] [27] [28].

4. Research Gaps in AI Dependence and Cognitive Impact

Causal Evidence and Underrepresented Populations

- **Longitudinal Gaps:** There is a pronounced lack of longitudinal and experimental research establishing causal effects of AI dependence on cognition [3] [39] [42].
- **Diversity Deficit:** Early learners, marginalized groups, and non-STEM disciplines are underexplored, limiting generalizability [39] [42].

Missing Theoretical Frameworks

- **Neuropsychological Models:** Few studies integrate neuropsychological frameworks addressing cognitive offloading, automation bias, and neuroplasticity [5] [43].
- **Quantum-Like Cognition:** Quantum-like models offer promise for unifying human and AI cognition but remain largely theoretical [43].
- **Socio-Technical Theories:** Socio-technical and organizational learning theories are needed to explain human-AI collaboration in real-world contexts [44].

Optimizing Pedagogical Strategies

- **Balance Needed:** Effective pedagogy requires inquiry-based learning, metacognitive training, and explicit AI literacy to balance AI assistance and cognitive engagement [45] [46] [47].
- **Ethical Agency:** Frameworks for critical agency and ethical AI use are underdeveloped, especially regarding academic integrity and algorithmic bias [48] [49].

Ethical Considerations and Human-AI Collaboration

- **Empirical Weakness:** Ethical frameworks and human-AI collaboration models are often theoretical, with limited empirical validation [44] [50] [51].
- **Transparency & Accountability:** There is a need for participatory, transparent, and accountable AI systems that preserve human agency [50] [51].

Synthesis: Addressing these gaps requires interdisciplinary, theory-driven, and inclusive research agendas [3] [39] [42].

5. AI Solutions Supporting Learning Without Replacing Independent Thought

AI Tools and Their Effectiveness

- **Personalized Learning:** Tools like ChatGPT, Gemini, and intelligent tutoring systems enhance engagement and learning outcomes, especially when used as supplementary aids [3] [52] [53] [54].
- **Risks:** Unrestricted use can impair long-term retention and critical thinking; structured, scaffolded integration preserves cognitive engagement [3] [55].
- **Adaptive Scaffolding:** AI-driven scaffolding and feedback cycles support self-regulated learning and critical thinking, especially in language and STEM education [55] [56].

Supporting Marginalized and Underrepresented Populations

- **Critical Thinking Gains:** Generative AI can enhance critical thinking in marginalized groups when paired with adaptive scaffolding and inclusive pedagogy [57] [58].
- **Ethical Frameworks:** Relational autonomy and participatory design are essential for addressing power dynamics and ensuring equitable AI benefits [59].
- **Teacher Mediation:** Teachers play a pivotal role in promoting ethical, critical, and responsible AI use, especially for vulnerable learners [60] [61] [62].

Teacher Mediation and Academic Integrity

- **Mitigating Risks:** Teacher mediation is crucial for upholding academic integrity, fostering critical thinking, and preventing cognitive offloading [60] [61].
- **Professional Judgment:** Teachers act as ethical facilitators, promoting transparency, accountability, and reflective dialogue [62] [63].

AI and Collaborative Learning

- **Enhanced Collaboration:** AI tools can improve group coordination, communication, and engagement, but may reduce human interaction if overused [64] [65] [66].

- **Balanced Integration:** Human-centered, ethically guided AI use is necessary to maintain social interaction and holistic development [67] [68].

Synthesis: AI tools are most effective when designed and implemented to augment, not replace, independent thought—especially through adaptive scaffolding, teacher mediation, and ethical frameworks [3] [59] [60].

6. Design Principles for AI to Foster Cognitive Skills

Human-Centered AI Design

- **Cognitive Alignment:** AI should be designed for cognitive compatibility, load optimization, transparency, fairness, and collaborative intelligence [27] [69] [70] [71].
- **User Involvement:** Early and ongoing user involvement enhances trust and ensures AI meets human needs [70].

Meta-Cognitive Control Mechanisms

- **Scaffolding & Prompts:** Embedding metacognitive scaffolding, self-monitoring modules, and uncertainty-based human review triggers supports critical thinking and ethical compliance [72] [73] [74] [75].
- **Transparency:** Metacognitive control modules improve explainability, accountability, and adaptability in AI systems [75] [76].

Teaching AI Literacy for Augmentation

- **Frameworks:** AI literacy should emphasize augmentation, boundary awareness, and hybrid intelligence, preparing users to collaborate with AI rather than be replaced [77] [78] [79].
- **Critical Agency:** Training should foster critical evaluation, ethical awareness, and adaptive learning [77] [80].

AI in Design Thinking and Creativity

- **Creativity Support:** AI tools in design thinking education enhance creativity and critical thinking by supporting content creation, iterative design, and reflective practices—without fully offloading cognitive responsibility [81] [82] [83].
- **Pedagogical Scaffolding:** Intentional, critical, and pedagogically guided AI use is essential to sustain originality and flexibility [17] [83].

Metacognitive Scaffolding and Prompts

- **Critical Thinking Enhancement:** Generative AI as metacognitive scaffolding in flipped classrooms and inquiry-based learning significantly improves critical thinking, self-efficacy, and engagement [74] [84].
- **Prompt Engineering:** Metacognitive prompts foster deeper inquiry, reflection, and iterative learning, supporting responsible and critical AI use [85] [86] [87].

Synthesis: Human-centered, metacognitively scaffolded, and ethically designed AI systems are best positioned to foster cognitive skills and preserve intellectual autonomy [27] [72] [77] [81].

7. Conclusion and Recommendations

Summary of Findings

- AI dependence has a dual impact: it can enhance efficiency, creativity, and engagement, but also risks cognitive atrophy, memory decline, and reduced critical thinking if unreflectively used [1] [3] [27].
- Methodological limitations—especially the lack of longitudinal, neuroimaging, and cross-cultural studies—impede robust causal inference.
- Effective solutions include metacognitive scaffolding, teacher mediation, adaptive feedback, and human-centered design.

Recommendations for AI Design and Pedagogy

- **Design for Augmentation:** Prioritize cognitive alignment, transparency, and fairness in AI systems [47].
- **Scaffolded Integration:** Use metacognitive prompts, adaptive scaffolding, and teacher mediation to foster critical thinking and self-regulation [60] [88].
- **Ethical Frameworks:** Implement participatory, relational, and transparent ethical frameworks, especially for marginalized populations [59].
- **AI Literacy:** Develop curricula that emphasize AI literacy, critical agency, and hybrid intelligence [77] [78].

Future Research Directions

- **Longitudinal Studies:** Invest in theory-driven, longitudinal, and interdisciplinary research to establish causal links and neural mechanisms [3] [43].
- **Inclusive Research:** Expand studies to underrepresented groups, early learners, and non-STEM disciplines.
- **Unified Frameworks:** Develop and empirically validate neuropsychological, quantum-like, and socio-technical frameworks for AI and cognition.

In summary: The next generation of AI systems must be designed not as cognitive substitutes, but as partners that scaffold, augment, and empower human thought. This requires a concerted effort across research, design, and pedagogy to ensure that AI serves as a catalyst for cognitive growth, ethical reasoning, and intellectual autonomy [1] [2] [3].

References

1. AI-Induced Cognitive Atrophy: Experimental, Neuroplastic, and Evolutionary Perspectives Venugopalachar, S., Chinnasamy, A., Vijayalakshmi, K. Psychological and Neurological Consequences of AI: Dependency, Laziness, and Reduced Critical Thinking, 2026 https://www.scopus.com/pages/publications/105047762536?origin=scopusAI

2. Policy guidelines and recommendations on AI use in teaching and learning: A meta-synthesis study Funa, A.A., Gabay, R.A.E. Social Sciences and Humanities Open, 2025

https://www.scopus.com/pages/publications/85209112463?origin=scopusAI

3. Working memory and generative AI tools in higher education: A systematic review Almudara, S.B., Alshehri, O.A., Alzahrani, S.Y., (...), Ibrahim, A.R. Research Journal in Advanced Humanities, 2026 https://www.scopus.com/pages/publications/105036063785?origin=scopusAI

4. Neuropsychological Effects of Overreliance on Artificial Intelligence Vieira, F. Psychological and Neurological Consequences of AI: Dependency, Laziness, and Reduced Critical Thinking, 2026 https://www.scopus.com/pages/publications/105047765521?origin=scopusAI

5. Cognitive Offloading to Cognitive Atrophy: A Neuropsychological Framework of AI Dependency, Automation Bias, and the Erosion of Critical Thinking Gill, A.A., Janjua, F., Zahoor, M., Afzal, A. Psychological and Neurological Consequences of AI: Dependency, Laziness, and Reduced Critical Thinking, 2026 https://www.scopus.com/pages/publications/105047759507?origin=scopusAI

6. Neuroplastic Consequences of Artificial Intelligence Dependency: Effects on Executive Networks and Cognitive Control Britel, I., Moummou, H., Aghdou, Z., (...), El-Mansoury, B. Psychological and Neurological Consequences of AI: Dependency, Laziness, and Reduced Critical Thinking, 2026 https://www.scopus.com/pages/publications/105047760653?origin=scopusAI

7. Your brain on ChatGPT, but whose brain? The missing adolescent in AI-cognition research Pereira Campos, J., Koff, T. Frontiers in Developmental Psychology, 2026 https://www.scopus.com/pages/publications/105048666933?origin=scopusAI

8. A model of bi-directional interactions between complementary learning systems for memory consolidation of sequential experiences Howard, M.D., Skorheim, S.W., Pilly, P.K. Frontiers in Systems Neuroscience, 2022 https://www.scopus.com/pages/publications/85140756158?origin=scopusAI

9. Exploring AI's Influence on Human Thinking: Productivity, Learning, and Cognitive Skills in the Age of XAI Rahulkumar, P.B., Hasmukhbhai, P.J., Himanshu, P.K., (...), Trivedi, H. 2025 Artificial Intelligence and Smart Technologies for Sustainability Conference, AISTS 2025, 2025 https://www.scopus.com/pages/publications/105030923100?origin=scopusAI

10. The mediating digital literacy and the moderating role of academic support in the relationship between artificial intelligence usage and creative thinking in nursing students Agaoglu, F.O., Bas, M., Tarsuslu, S., (...), Agaoglu, N.B. BMC Nursing, 2025 https://www.scopus.com/pages/publications/105004013063?origin=scopusAI

11. The Impact of Generative AI Dependence on University Students' Creativity: The Mediating Role of Critical Thinking and the Moderating Role of Thinking Orientation Yuan, F., Liu, C., Yang, J. Journal of Intelligence, 2026 https://www.scopus.com/pages/publications/105048596859?origin=scopusAI

12. The mediating effects of critical thinking on the motivation and creativity of Business English learners in the age of AI: Cognitive flexibility theory Lijie, H., Kun, T., Leng, C.H., Salleh, U.K.B.M. Thinking Skills and Creativity, 2024 https://www.scopus.com/pages/publications/85197532745?origin=scopusAI

13. The impact of artificial intelligence on students' 4C skills: A meta-analysis Tian, Q., Zheng, X. Educational Research Review, 2025 https://www.scopus.com/pages/publications/105015085809?origin=scopusAI

14. The interplay between creativity, thinking styles, higher education, and generative AI Fields, Z. Impacts of Generative AI on Creativity in Higher Education, 2024 https://www.scopus.com/pages/publications/85207550714?origin=scopusAI

15. Learners' AI dependence and critical thinking: The psychological mechanism of fatigue and the social buffering role of AI literacy Tian, J., Zhang, R. Acta Psychologica, 2025 https://www.scopus.com/pages/publications/105018241598?origin=scopusAI

16. AI Dependence and Self-Reported Creativity Among Chinese College Students: The Roles of Motivation for AI Use and Perceived AI Literacy Hou, J., Huang, Y., Wang, Q., (...), Zhao, F. Behavioral Sciences, 2026

https://www.scopus.com/pages/publications/105051438459?origin=scopusAI

17. AI-Enabled Creativity: Effects on Divergent and Convergent Thinking among Higher Education Students Karthikeyan, K., Vasudevan, A., Sowmya, K.R., (...), Mohamad Hanefar, S.B. International Journal of Engineering Pedagogy, 2025 https://www.scopus.com/pages/publications/105024448455?origin=scopusAI

18. Heuristic or Algorithmic Thinking? Assessing Generative AI's Impact on Human Decision-Making. A Literature Review Fernández-Concha, R., Solórzano Muñante, M.D.C. Applied Artificial Intelligence, 2026 https://www.scopus.com/pages/publications/105051316172?origin=scopusAI

19. Smart Tech Roulette: Insights for Exploring AI's Impact on Learning, Cognitive Development, and Ethics in Context Hoke, T. Lecture Notes in Networks and Systems, 2025 https://www.scopus.com/pages/publications/105021832933?origin=scopusAI

20. The Impact of Culture on Human-AI Collaboration in a Decision-Making Task: A Comparison Between North Americans and South Asians Schwanz, R., Fox, E.L., Lui, T., (...), Capiola, A. International Journal of Human-Computer Interaction, 2026 https://www.scopus.com/pages/publications/105032886103?origin=scopusAI

21. Executive functions as predictors of critical thinking: Behavioral and neural evidence Li, S., Ren, X., Schweizer, K., (...), Wang, T. Learning and Instruction, 2021 https://www.scopus.com/pages/publications/85089189760?origin=scopusAI

22. Generative artificial intelligence reliance and executive function attenuation: Behavioral evidence of cognitive offload in high-use adults. Baldeo, S. Technology, Mind, and Behavior, 2026 https://www.scopus.com/pages/publications/105051947429?origin=scopusAI

23. A cross-national assessment on critical thinking disposition between Hungarian and Chinese undergraduate students Liu, Y., Pásztor, A., Molnár, G. Thinking Skills and Creativity, 2026 https://www.scopus.com/pages/publications/105026150112?origin=scopusAI

24. Reinforcing L2 reading comprehension through artificial intelligence intervention: refining engagement to foster self-regulated learning Shafiee Rad, H. Smart Learning Environments, 2025 https://www.scopus.com/pages/publications/86000070834?origin=scopusAI

25. How Does Self-Regulated Learning Resist AI Dependence? A Mediating Effect Study Based on College Students Who Frequently Use AIGC Tools Liu, B. SAGE Open, 2026 https://www.scopus.com/pages/publications/105030839303?origin=scopusAI

26. AI-Driven Interventions for Neurocognitive, Self-Regulation, and Adaptive Skill Development in Neurodevelopmental and Cognitive Disorders: A Systematic Review of Randomized Controlled Trials Mitsea, E., Drigas, A., Skianis, C. Healthcare (Switzerland), 2026 https://www.scopus.com/pages/publications/105051612408?origin=scopusAI

27. A systematic review of meta-cognitive approaches to AGI Bohra, N.M., Wagle, S.A., Patil, S. Discover Artificial Intelligence, 2026 https://www.scopus.com/pages/publications/105043995688?origin=scopusAI

28. Epistemic Agency and Critical Subjectivity in the age of Artificial Intelligence Conde, S.D., Pesenti, L., Rodriguez, J. Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology, 2026 https://www.scopus.com/pages/publications/105048862643?origin=scopusAI

29. The Untapped Potential of Dimension Reduction in Neuroimaging: Artificial Intelligence-Driven Multimodal Analysis of Long COVID Fatigue Rudroff, T., Klén, R., Rainio, O., Tuulari, J. Brain Sciences, 2024 https://www.scopus.com/pages/publications/85213472195?origin=scopusAI

30. THE IMPACT OF AI-BASED FEEDBACK ON METACOGNITIVE SKILLS OF UNIVERSITY STUDENTS: A CROSS-SECTIONAL ANALYSIS Chirinos, R.A.G., Benítez, H.A.P., Paniagua, P.J.V., (...), Blacksmith, M.C.R. Acta Innovations, 2026 https://www.scopus.com/pages/publications/105044090685?origin=scopusAI

31. AI, Metacognition, and the Verification Bottleneck: A Three-Wave Longitudinal Study of Human Problem-Solving Huemmer, M., Durner, F., Shyiramunda, T., Cummings-Koether, M.J. 2nd International Conference on Human-AI Interaction and Experience Design, HAXD 2026, 2026 https://www.scopus.com/pages/publications/105047240556?origin=scopusAI

32. An Introduction to Single-Case Experimental Designs for Applied Human Factors and Ergonomics Laraway, S., Snycerski, S., Pradhan, S., (...), Battiste, V. Applied Human Factors and Ergonomics International, 2024 https://www.scopus.com/pages/publications/105031155894?origin=scopusAI

33. Deep-learning based multi-modal models for brain age, cognition and amyloid pathology prediction Wang, C., Zhang, W., Ni, M., (...), Gao, F. Alzheimer's Research and Therapy , 2025 https://www.scopus.com/pages/publications/105006930418?origin=scopusAI

34. A One-Year Longitudinal Study Examining the Direct and Indirect Effects of AI Dependence on Work Engagement and Gender Differences Wen, J., Lei, Y., Liu, Q. Behavioral Sciences, 2026 https://www.scopus.com/pages/publications/105028737527?origin=scopusAI

35. Causal inference using observational intensive care unit data: a scoping review and recommendations for future practice Smit, J.M., Krijthe, J.H., Kant, W.M.R., (...), van Genderen, M.E. npj Digital Medicine, 2023 https://www.scopus.com/pages/publications/85178347733?origin=scopusAI

36. Evaluating generative AI-assisted inquiry-based learning and metacognitive scaffolding for enhancing scientific vocabulary, content comprehension, and lab report writing Lai, C.-J. Interactive Learning Environments, 2026 https://www.scopus.com/pages/publications/105019595603?origin=scopusAI

37. Leveraging Deep Reinforcement Learning for Metacognitive Interventions Across Intelligent Tutoring Systems Abdelshiheed, M., Hostetter, J.W., Barnes, T., Chi, M. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2023 https://www.scopus.com/pages/publications/85159875984?origin=scopusAI

38. A Systematic Literature Review of Agentic AI: Definitions, Architectures, and Challenges Moralles, C., Costa, L.A.L.F.D., Rigo, S.J., (...), Roehrs, A. IEEE Access, 2026 https://www.scopus.com/pages/publications/105031511266?origin=scopusAI

39. Empirical Research Status of Generative AI in K-12 Education: A Systematic Literature Review Liao, Y., Zhou, J., Xu, G. 8th International Conference on Educational Technology Management, ICETM 2025, 2025 https://www.scopus.com/pages/publications/105035373588?origin=scopusAI

40. Bias and ethics of AI systems applied in auditing - A systematic review Murikah, W., Nthenge, J.K., Musyoka, F.M. Scientific African, 2024 https://www.scopus.com/pages/publications/85196552319?origin=scopusAI

41. Learning AI Auditing: A Case Study of Teenagers Auditing a Generative AI Model Morales-Navarro, L., Gan, M., Yu, E., (...), Metaxa, D. Proceedings of the ACM on Human-Computer Interaction, 2025 https://www.scopus.com/pages/publications/105019391318?origin=scopusAI

42. The Global Landscape of Research on AI Literacy: A Bibliometric Analysis of Trends, Collaborations, and Emerging Themes Kumar, J.A. Proceedings - Frontiers in Education Conference, FIE, 2025 https://www.scopus.com/pages/publications/105033056318?origin=scopusAI

43. Quantum-like cognition and AI toward the convergence of natural and artificial intelligence Khrennikov, A. Discover Artificial Intelligence, 2026 https://www.scopus.com/pages/publications/105030823845?origin=scopusAI

44. Human-AI Cognitive Collaboration in Industrial Contexts: A Managerial Framework for Assessment, Execution, and Governance-From Theory to Sustainable Practice Patnaik, T., Nanda, N.S. Human-AI Cognitive Collaboration for Industrial Evolution, 2026 https://www.scopus.com/pages/publications/105043571133?origin=scopusAI

45. Examining Teaching Competencies and Challenges While Integrating Artificial Intelligence in Higher Education Ren, X., Wu, M.L. TechTrends, 2025 https://www.scopus.com/pages/publications/85218181321?origin=scopusAI

46. Beyond AI Dependence: Pedagogical Approaches to Strengthen Student Reasoning and Analytical Skills Shaikh, S.A., Jagirdar, A.H. Empowering Learners With AI: Strategies, Ethics, and Frameworks, 2025 https://www.scopus.com/pages/publications/105025187975?origin=scopusAI

47. CARE+: A Methodology for AI-Enhanced Learning that Develops 21st Century Competencies Cárdenas Toledo, C., Soto Guerrero, F.M., Loaiza, M.I. Communications in Computer and Information Science, 2026 https://www.scopus.com/pages/publications/105020719395?origin=scopusAI

48. From Algorithmic Output to Human Understanding: Building Critical Agency and Trust in AI-Mediated Research and Education Saha, S., Singh, M.K., Choudhary, P.K. Epistemic Trust in AI-Generated Knowledge: Global Trends in Research Ethics, 2026 https://www.scopus.com/pages/publications/105049985356?origin=scopusAI

49. Examining the Influence of AI Dependency, Ethical Awareness, and Algorithmic Bias Awareness on Critical Thinking: Nudging the Algorithmic Mind Mishra, A. Psychological and Neurological Consequences of AI: Dependency, Laziness, and Reduced Critical Thinking, 2026 https://www.scopus.com/pages/publications/105047763454?origin=scopusAI

50. Ethical Considerations in the Integration of Artificial Intelligence: A Systematic Review Geçikli, M. Navigating Modern Digital Communication Ethics and Law, 2025 https://www.scopus.com/pages/publications/105017840717?origin=scopusAI

51. The ethics of AI–human collaboration: a socio-technical framework for responsible governance in complex global operations Sposato, M., Dittmar, E.C. Journal of Information, Communication and Ethics in Society, 2026 https://www.scopus.com/pages/publications/105046509167?origin=scopusAI

52. The impact of artificial intelligence-based learning tools in academic innovation: a review of Deep seek, GPT, and Gemini (2020–2025) Younas, M., El-Dakhs, D.A.S., Noor, U. Frontiers in Education, 2025 https://www.scopus.com/pages/publications/105023677812?origin=scopusAI

53. The Influence of Artificial Intelligence Tools on Learning Outcomes in Computer Programming: A Systematic Review and Meta-Analysis Alanazi, M., Soh, B., Samra, H., Li, A. Computers, 2025 https://www.scopus.com/pages/publications/105006728473?origin=scopusAI

54. A Systematic Review of Artificial Intelligence Integration in Digital Learning for Higher Education: Technologies, Personalization, and Implementation Challenges Michelle, Miranda, E. Proceedings of 2025 11th International HCI and UX Conference in Indonesia, CHIuXiD 2025, 2025 https://www.scopus.com/pages/publications/105033055479?origin=scopusAI

55. Generative AI and its Applications to Education: A Systematic Literature Review Shadiev, R., Kairatkyzy, D., Mamyr, S., (...), Bekezhanova, A. 2026 7th International Conference on Information Technology and Education Technology, ITET 2026, 2026 https://www.scopus.com/pages/publications/105048759869?origin=scopusAI

56. Artificial Intelligence as a Mediator of Scaffolding: A Scoping Review Tenisheva, K.A., Litvinova, A.I. Ojkumena Regional Researches, 2025 https://www.scopus.com/pages/publications/105018642117?origin=scopusAI

57. Toward sustainable aesthetic education: exploring the impact of GenAl collaboration on students' critical thinking disposition Liu, H., Niu, T., Shengyan, C., (...), Taclis Luo, Y. Frontiers in Psychology, 2026 https://www.scopus.com/pages/publications/105044516462?origin=scopusAI

58. Leveraging artificial intelligence as a Vygotskian tool to enhance Filipino university students' language learning and critical thinking skills Ravelo, V.M., Ancho, I.V., Mecha, J.P.F., Albakri, I.S.M.A. Palawan Scientist, 2026 https://www.scopus.com/pages/publications/105043463345?origin=scopusAI

59. Decolonizing AI Ethics: Relational Autonomy as a Means to Counter AI Harms Mhlambi, S., Tiribelli, S. Topoi, 2023 https://www.scopus.com/pages/publications/85147587043?origin=scopusAI

60. From AI assistance to pedagogical reflection: a rubric-mediated model for generative AI in teacher education Bilbao-Quintana, N., Tejada Garitano, E., Romero Andonegui, A., López de la Serna, A. Frontiers in Education, 2026 https://www.scopus.com/pages/publications/105044675782?origin=scopusAI

61. Beyond AI-Generated Text: Strengthening Academic Integrity and Critical Thinking in Higher Education Florencio da Silva, R. Information (Switzerland), 2026 https://www.scopus.com/pages/publications/105048328433?origin=scopusAI

62. Can Generative Artificial Intelligence be a Good Teaching Assistant?—An Empirical Analysis Based on Generative AI-Assisted Teaching Tang, Q., Deng, W., Huang, Y., (...), Zhang, H. Journal of Computer Assisted Learning, 2025 https://www.scopus.com/pages/publications/105004433950?origin=scopusAI

63. Generative Artificial Intelligence (AI) in the high school English classroom and its effect on students' identity and teachers' practice Riser, P. English in Education, 2025 https://www.scopus.com/pages/publications/105003156201?origin=scopusAI

64. The influence of artificial intelligence on collaborative learning dynamics in the classroom environment Du, Z., Dankina, G., Zhou, Z., (...), Karabalaeva, G. Social Sciences and Humanities Open, 2026 https://www.scopus.com/pages/publications/105038343141?origin=scopusAI

65. Rethinking Student-Robot Social Interactions in GenAI Capitalism and its Role in Shaping Cybersocialization through a Sociological Lens Al Yakin, A., Cardoso, L., Al Matar, A.S., (...), Hussein, M.W. Proceedings of International Conference on Applied Innovation in IT, 2025 https://www.scopus.com/pages/publications/105021949631?origin=scopusAI

66. SUPPORTING VIRTUAL LEARNING AND INTERNATIONAL ACADEMIC VISITS WITH AI Richter, T.T., Langesee, L.-M. IDIMT 2024: Changes to ICT, Management, and Business Processes through AI - 32nd Interdisciplinary Information Management Talks, 2024 https://www.scopus.com/pages/publications/85201793679?origin=scopusAI

67. Transformations in Learning and Social Relations in the Age of Artificial Intelligence: An Analysis From the Perspective of Education and Contemporary Culture Alesanco, E.B., Cuartas, R.S. AI Influence on Governance and Law in the Digital Age, 2026 https://www.scopus.com/pages/publications/105031047169?origin=scopusAI

68. Impact of Educational AI on Students' Studying Habits and Academic Performance Khalkho, R., Singh, S., Gupta, N., Srivastava, P. 2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Applications, ICAIQSA 2024 - Proceedings, 2024 https://www.scopus.com/pages/publications/105000118821?origin=scopusAI

69. Designing Human-Centered and Cognitively Aligned Artificial Intelligence Systems for Ethical and Effective Decision-Making Kumari, J., Yadav, P., Bhargava, C.P. Brain-Inspired Cognitive Mechanisms for AIEnhanced Decision Intelligence, 2026 https://www.scopus.com/pages/publications/105048157049?origin=scopusAI

70. Framing the Human-Centered Artificial Intelligence Concepts and Methods: Scoping Review Bevilacqua, R., Bailoni, T., Maranesi, E., (...), Sakellariou, G. JMIR Human Factors, 2025 https://www.scopus.com/pages/publications/105007924954?origin=scopusAI

71. Beyond Assistance: Embracing AI as a Collaborative Co-Agent in Education Katsenou, R., Kotsidis, K., Papadopoulou, A., (...), Deliyannis, I. Education Sciences, 2025 https://www.scopus.com/pages/publications/105014419796?origin=scopusAI

72. Reconceptualizing AI Literacy: The Importance of Metacognitive Thinking in an Artificial Intelligence (AI)-Enabled Workforce Sidra, S., Mason, C. Proceedings - 2024 IEEE Conference on Artificial Intelligence, CAI 2024, 2024 https://www.scopus.com/pages/publications/85201216512?origin=scopusAI

73. Mitigating cognitive offloading in GenAI-supported collaboration: a metacognitive scaffolding approach to enhancing students' critical thinking Zhuo, R. Interactive Learning Environments, 2026 https://www.scopus.com/pages/publications/105045070608?origin=scopusAI

74. GenAI as metacognitive scaffolding in flipped classrooms: Impact on learner outcomes Yildiz Durak, H., Kaya, D., Ursavaş, Ö.F. Innovations in Education and Teaching International, 2025 https://www.scopus.com/pages/publications/105019340081?origin=scopusAI

75. Metacognitive AI Wei, H., Shakarian, P., Lebiere, C., (...), Nirenburg, S. Metacognitive Artificial Intelligence, 2025 https://www.scopus.com/pages/publications/105035565759?origin=scopusAI

76. Harnessing Metacognition for Safe and Responsible AI Walker, P.B., Haase, J.J., Mehalick, M.L., (...), Davidson, I.N. Technologies, 2025 https://www.scopus.com/pages/publications/105001138503?origin=scopusAI

77. Prioritizing the core dimensions of teacher AI literacy for AI-empowered teaching capability in higher education: a Delphi study Li, H., Suo, Z., Liu, C., (...), Wider, W. Interactive Learning Environments, 2026 https://www.scopus.com/pages/publications/105044888971?origin=scopusAI

78. Conceptualising Human–AI Collaboration Competency for hybrid intelligence: towards learner agency through a metacognitive perspective Dang, B., Hong, Y., Doyle, C., Nguyen, A. Studies in Higher Education, 2026 https://www.scopus.com/pages/publications/105052039476?origin=scopusAI

79. Navigating the Impact of Augmentation Computing Intelligence, AI Literacy, and Human Integration in English Language Education 7.0 by Using Computer-Assisted Language Learning and Augmented (CALLedA) Model Muthmainnah, M., Cardoso, L., Al Matari, A.S., (...), Susilo, J. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105027028398?origin=scopusAI

80. The Impact of AI Literacy Training on Engineering Creativity: An Empirical Study Li, H., Deng, M., Yang, Y., Fu, M. 2026 8th International Conference on Computer Science and Technologies in Education, CSTE 2026, 2026 https://www.scopus.com/pages/publications/105048965389?origin=scopusAI

81. Using Al to Foster Creativity, Critical Thinking, and Innovation in Students Suntharalingam, P. Intelligent Learning Tools, Enhancing Student Skills, and Career Preparedness, 2026 https://www.scopus.com/pages/publications/105033568075?origin=scopusAI

82. Promoting the Development of Critical Thinking Among Design Novices in AIGC-Aided Design Processes Ge, P., Fan, F. Proceedings of DRS, 2026 https://www.scopus.com/pages/publications/105046354331?origin=scopusAI

83. Generative AI in Design Thinking Pedagogy: Enhancing Creativity, Critical Thinking, and Ethical Reasoning in Higher Education Rana, V., Verhoeven, B., Sharma, M. Journal of University Teaching and Learning Practice , 2025 https://www.scopus.com/pages/publications/105016471796?origin=scopusAI

84. Adaptive vs. planned metacognitive scaffolding for computational thinking: Evidence from generative AI-supported programming in elementary education Liu, J., Zhang, Y., Li, W., (...), Zhang, X. Computers and Education, 2026 https://www.scopus.com/pages/publications/105018233693?origin=scopusAI

85. Exploring metacognitive regulation activities and their roles in prompt literacy development among humanities and social science researchers in GenAI Jing, Y., Zhao, Y., Wang, D., (...), Zhu, Q. Information Research, 2026 https://www.scopus.com/pages/publications/105034287333?origin=scopusAI

86. Impact of the Integration of Metacognitive Prompts by Generative Artificial Intelligence (GenAI) in Collaborative and Individual Learning in Improving Writing Skills and Metacognitive Awareness Haidar, H., Suryoputro, G., Safi'i, I. International Journal of Learning, Teaching and Educational Research, 2025 https://www.scopus.com/pages/publications/105009621205?origin=scopusAI

87. MetaC Framework: Metacognitive Prompting for AI-Driven Coursework Assessment in Tertiary Education Sinnappan, G.S., Shauki, N.B.I., Cheng, W.F., Teh, L.W. 2025 17th International Conference on Knowledge and

Smart Technology, KST 2025, 2025 https://www.scopus.com/pages/publications/105007550659?origin=scopusAI

88. Cognitive Dependency and Pedagogical Resilience in AISaturated Classrooms: Critical Thinking in the Age of AI Alam, A. Psychological and Neurological Consequences of AI: Dependency, Laziness, and Reduced Critical Thinking, 2026 https://www.scopus.com/pages/publications/105047762598?origin=scopusAI